

Test Yourself

Microeconomics

Price Elasticity of

Demand

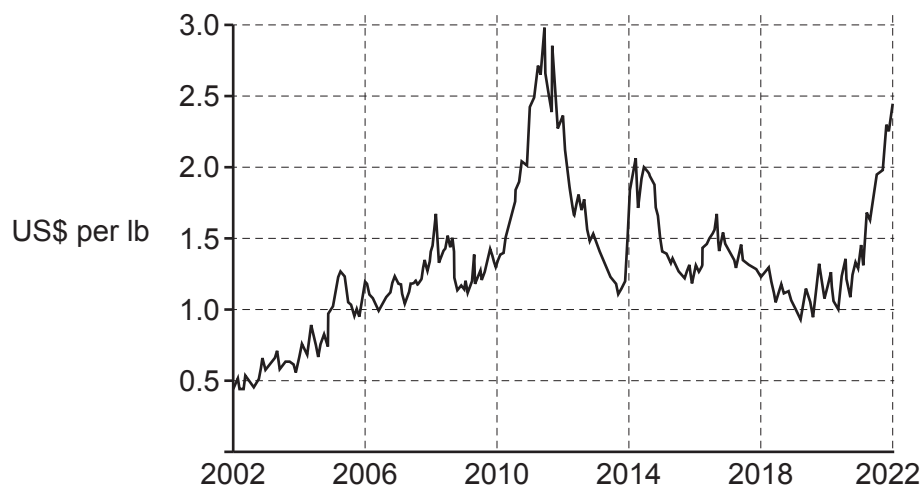
Section A

Case Study 1

Global coffee bean prices reach a new high

23/O/N/24

Coffee is big business, especially in the United States (US). The global market was worth nearly US\$110bn in 2020, with production around 10m tonnes of coffee beans. 95% of this production came from thousands of small-scale producers in South America, Central America, Asia and Africa. If present trends continue, production is forecast to triple by 2050. The Arabica coffee bean price was at a 10-year high in January 2022, having more than doubled in 2021. Fig. 1.1 shows the world price of coffee in United States dollars (US\$) per pound (lb) weight. A pound is 454 grammes.



Source: *tradingeconomics.com*, 18 February 2022

Fig. 1.1: Arabica coffee bean price, 2002 to 2022

So how can this huge price increase be explained? The main cause was a series of weather events affecting Brazil, the world's largest producer of high quality Arabica coffee beans. Its market share is 35% of total global supply. There was severe drought in early 2021 that reduced the number of 'cherries', which contain the beans on coffee bushes. The crop yield was further damaged by frosts that followed the drought. As a result of these weather events, Brazil produced its smallest volume of quality coffee beans for ten years.

In addition, the production of cheaper low-quality Robusta coffee beans in Vietnam was badly affected by storms which stripped the bushes of their 'cherries'. Overall, there was a very large decrease in the global supply of all coffee beans.

Supply issues affecting major producing countries like Brazil and Vietnam mean that the volume of coffee beans produced regularly fluctuates between 'high' years and 'low' years. Producers try to reduce their risks through buffer stock schemes in order to maintain a regular income stream. This type of scheme is particularly important for the many small-scale subsistence producers who have no other source of income.

The dramatic weather events of 2021 have been exceptional, although a few large-scale producers in Brazil who survived the immediate impact of drought and frost should gain from the huge rise in the price of coffee beans. But what about the thousands of other producers elsewhere who lack power in the market? The most likely outcome is that they will once again become victims of the unpredictable global market for coffee beans.

Case Study 2

Dilemma for the European Central Bank (ECB)

22/M/J/24

The Eurozone consists of 19 European Union (EU) states that have agreed to use a common currency (the euro) and monetary policy, both of which are governed by the ECB. The ECB sets interest rates for all 19 members of the Eurozone and since July 2021 has aimed to maintain the rate of inflation at 2% in the medium term. Interest rates have been held at 0% since March 2016.

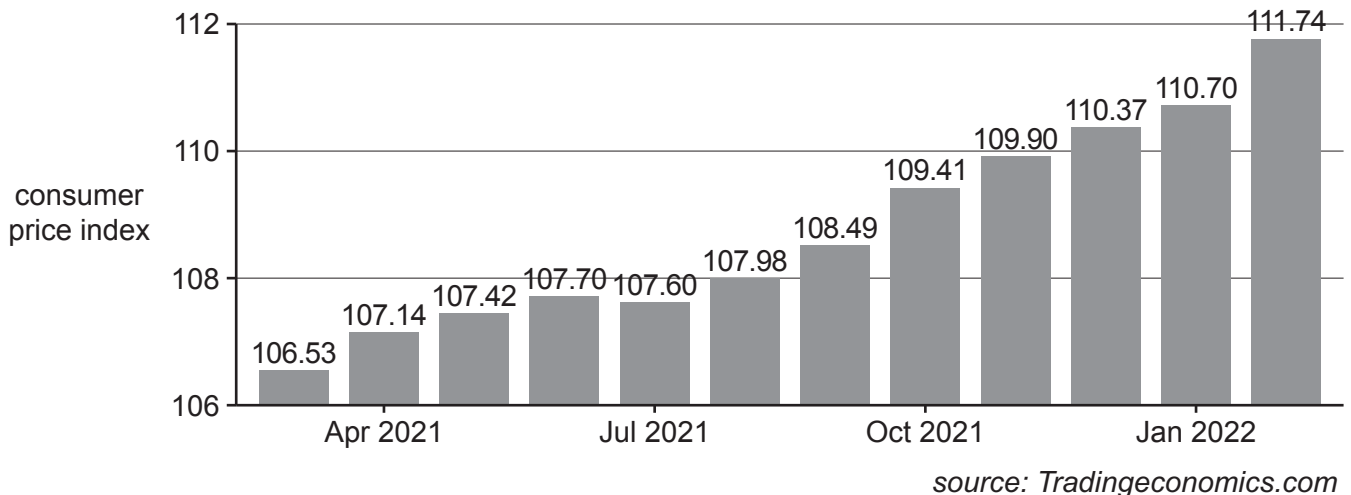


Fig 1.1: Eurozone Consumer Prices Index (CPI) March 2021 – February 2022

In March 2022, the annual rate of inflation hit a record high for the Eurozone of 7.5% and is forecast to continue to rise throughout 2022. Additionally, according to the ECB vice president, annual economic growth is expected to fall from 4.6% at the end of 2021 to around 0% at the end of 2022. Most of the impact of these changes is expected to fall on consumers. Much higher energy costs and rising food prices tend to have a more severe effect on poorer households and those on fixed incomes.

The main causes of the rapid increase in the inflation rate are supply-side factors. The rise in energy prices result from a combination of the Covid-19 pandemic and the conflict between Russia and Ukraine which have reduced supplies of oil and gas. Additionally, the Eurozone labour market is increasingly suffering from a shortage of supply as unemployment has fallen to a record low of 6.8% in February 2022 with further falls predicted. Because the rise in the rate of inflation is almost exclusively supply-side driven, the ECB fears that this will lead to further falls in economic growth leading to a period of 'stagflation', where an economy experiences high inflation and low economic growth at the same time.

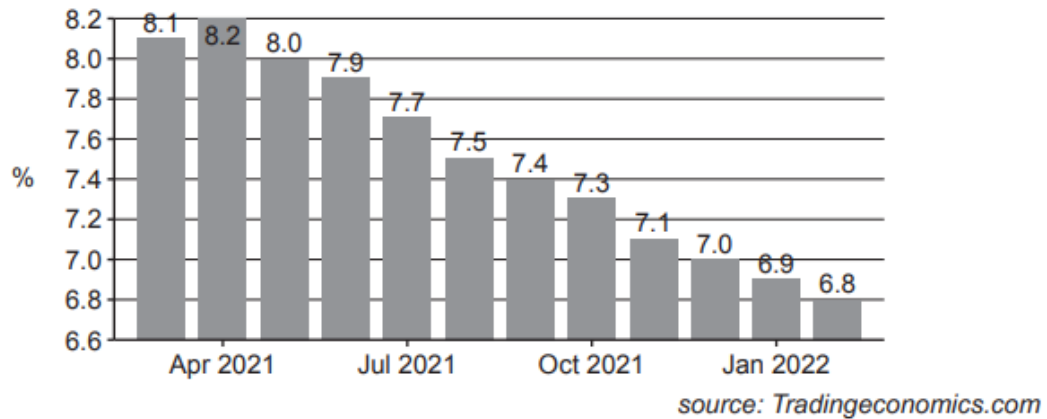


Fig 1.2: Eurozone % unemployment rate March 2021 – February 2022

All this leaves the ECB with a dilemma. Should it:

- increase interest rates substantially now to control the increasing rate of inflation and risk weakening economic growth even further, or
- increase them slightly in the hope that supply pressure will ease soon (this runs the risk of making high inflation more permanent if the pressure does not ease), or
- leave interest rates unchanged?

Case study 1

1. Assess the extent to which coffee bean producers will gain from the huge increase in coffee bean prices in 2021. [6]

Case study 2

2. Using the concept of price elasticity of demand, assess the relative impact on poorer households, including those on fixed incomes, of rising prices of food and energy. [6]

Section B

Duration of each question is **40 mins**

1. With the help of a formula, explain what is meant by the term price elasticity of demand for a product **and** consider the extent to which it allows an entrepreneur to determine the effect of price changes on the total expenditure on their product. (8) **q2a/22/O/N/23**

Section C

Duration 40 mins

1. The table shows Lee's and Yim's price elasticity of demand for restaurant meals and cinema tickets.

	Lee	Yim
restaurant meals	-1.2	-0.8
cinema tickets	-0.7	-1.3

There is a rise in the price of restaurant meals and a fall in the price of cinema tickets.

What can be concluded after these price changes?

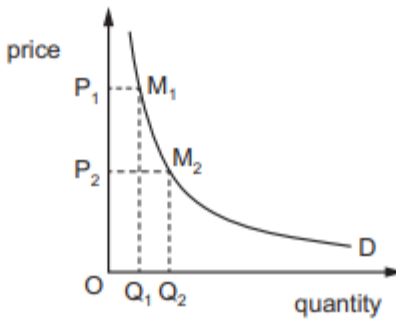
- A** Restaurant owners will receive more income.
- B** Lee will spend more money on both cinema tickets and restaurant meals.
- C** Yim will spend more money on both cinema tickets and restaurant meals.
- D** Cinema owners will receive less income.
2. A sports stadium has two prices of seats. The cheaper seats are \$10 and the more expensive seats are \$20. When the prices of all seats are increased by 10%, the effects on demand are a fall of 5% for the cheaper seats and a fall of 12% for the more expensive seats.

What describes the responsiveness of demand to the change in price?

	cheaper seats	more expensive seats
A	elastic	elastic
B	elastic	inelastic
C	inelastic	elastic
D	inelastic	unitary elasticity

3. In which situation is the demand for a product said to be price elastic?
- A** The quantity demanded responds to a change in price.
- B** An increase in price brings about a decrease in the quantity demanded.
- C** An increase in price induces consumers to spend more on the product.
- D** A decrease in price brings about an increase in revenue.

4. In the diagram, area $OP_1M_1Q_1$ is equal to area $OP_2M_2Q_2$.



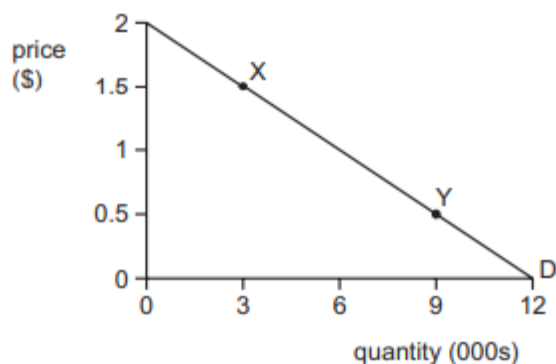
What is the value of the price elasticity of demand if the price is halved from P_1 to P_2 ?

- A -1
 - B -0.5
 - C zero
 - D infinity
5. When is the price elasticity of demand for a product most likely to be price elastic?
- A A large percentage of income is spent on the product.
 - B It is measured in the short run.
 - C It is a necessity product.
 - D There are few substitutes available.
6. When the price of a firm's product falls by 5%, its total revenue also falls by 5%.

What describes the product's price elasticity of demand?

- A elastic
- B infinite
- C unitary
- D zero

7. The diagram shows a demand curve for pineapples.



What happens to the value of price elasticity of demand (PED) when there is a movement from point X to point Y and what describes the value of PED at point Y?

	effect on value of PED	value of PED at Y
A	fall	elastic
B	fall	inelastic
C	rise	elastic
D	unchanged	unitary

8. The table shows the demand schedule for textbooks.

price \$	quantity demanded
50	0
40	100
30	200
20	300
10	400
0	500

What can be concluded from this information?

- A** Profit would be maximised at a price of \$30.
- B** Profit would not change if the price rose from \$20 to \$30.
- C** Revenue would increase if the price was decreased from \$40 to \$30.
- D** Revenue would be maximised at a price of \$40.

9. The economics department of a university calculates that the price elasticity of demand for the products of a local firm is between -1.1 and -1.3 .

Which decision could be justified by the research?

- A Increase prices to increase profit.
- B Invest in more capital to increase output.
- C Reduce advertising to reduce costs.
- D Reduce prices to increase revenue.

10. A manufacturer progressively reduces the price of his product.

The table shows the outcome of this policy.

price (\$)	total revenue 000s (\$)
10	750
9	750
8	750

What is the price elasticity of demand for the product?

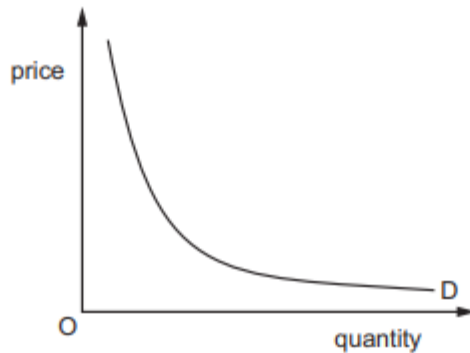
- A perfectly inelastic
- B relatively inelastic
- C perfectly elastic
- D unitary

11. In the UK, attempts to encourage people to change from road to rail travel by the introduction of a system of road pricing were forecast to have little effect because 'people like using their cars too much'.

How would this forecast be explained?

- A The price elasticity of demand for cars is high.
- B The price elasticity of demand for petrol is high.
- C The price elasticity of demand for rail travel is low.
- D The price elasticity of demand for car travel is low.

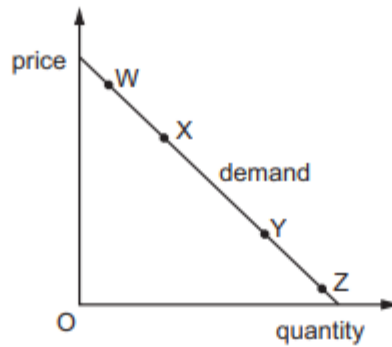
12. The diagram shows the demand curve for a product with unitary price elasticity.



What will happen with such a curve?

- A A fall in price will bring about an increase in total expenditure on the product.
 - B A fall in price will bring about an increase in sales but a fall in total expenditure on the product.
 - C As the price rises, total expenditure on the product will stay the same.
 - D As the price rises, total expenditure on the product will rise and then fall.
13. Why is the price elasticity of demand for a particular brand of coffee likely to be higher than the price elasticity of demand for coffee in general?
- A Coffee in general takes up a higher proportion of consumers' income.
 - B Consumers may be very loyal to a particular brand of coffee.
 - C There are more substitutes for a particular brand of coffee than for coffee in general.
 - D Total demand for coffee is higher than for any one brand.
14. What would be the most likely reason for a firm's decision to calculate the price elasticity of demand for its product?
- A to provide retail statistics for the government
 - B to predict the change in its total revenue after it raises prices
 - C to calculate the firm's competitive position
 - D to enable the firm to complete consumer satisfaction surveys

15. The diagram shows the demand curve for a normal product.



Which two points indicate a move from a price inelastic point to a less price inelastic point?

- A** W to X **B** X to W **C** Y to Z **D** Z to Y

16. Which product has the lowest price elasticity of demand?

	price rises by (%)	quantity demanded falls by (%)
A	0	10
B	5	5
C	10	5
D	10	0

17. Over the last ten years the price elasticity of demand for tea in many countries has risen.

What is the most likely cause of this change in price elasticity?

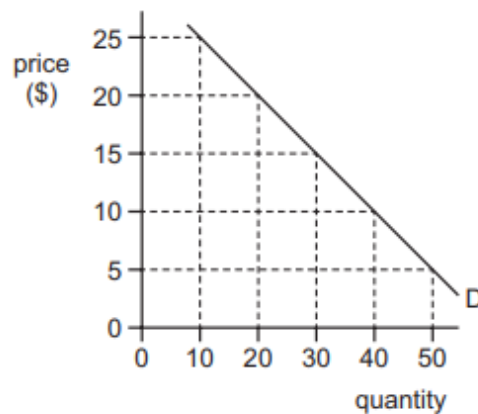
- A** a decrease in the incomes of consumers
B a decrease in the number of complements to tea
C an increase in the amount of tea supplied
D an increase in the number of substitutes for tea

18. The curve in the diagram shows a relationship between the price and the quantity of a product. It has not been given a label.



What is an accurate description of the curve?

- A a perfectly elastic demand curve
 - B a perfectly inelastic supply curve
 - C a relatively elastic supply curve
 - D a unitary elastic demand curve
19. The diagram shows the demand for a product for which there are only three buyers, Freeman, Hardy and Willis.



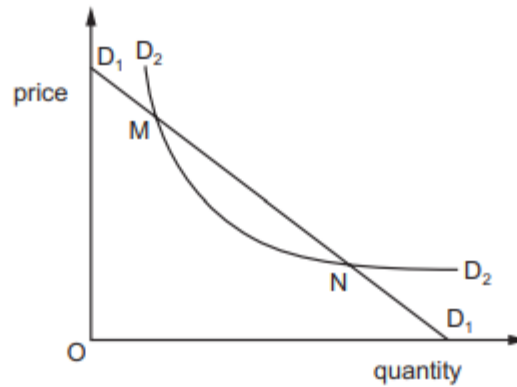
The table shows the demand from two of the three buyers.

price (\$)	Freeman's demand	Hardy's demand
10	24	6
15	15	5
20	6	4

Which statement about the demand for the product from Willis is correct?

- A It has unitary price elasticity of demand.
- B It is a downward-sloping straight-line demand curve.
- C It is a perfectly elastic demand curve.
- D It is a perfectly inelastic demand curve.

20. In the diagram D_1D_1 is a straight line demand curve and D_2D_2 is a rectangular hyperbola curve.



Which statement is correct?

- A D_2D_2 is more elastic than D_1D_1 at point M.
- B D_2D_2 is more elastic than D_1D_1 at point N.
- C The elasticity of demand increases on moving down both curves.
- D The elasticity of demand is inelastic at point M on both curves.

Marking Scheme

Section A (Case Study 1)

Question	Answer	Marks
1	Assess the extent to which coffee bean producers will gain from the huge increase in coffee bean prices in 2021. Up to 3 marks for explanation/analysis of the gains to coffee bean producers: • If the demand for coffee is price inelastic in demand, then coffee producers would be expected to gain (in terms of increased revenue). • For example, increased revenue per lb. of coffee sold; increased funds available to invest and innovate in better methods of storing coffee beans in the event of future storms. • Producers in Brazil and Vietnam and with existing stocks of coffee that survived the frost and storms would expect to increase their own sales at the expense of those rivals unable to produce a large enough yield of coffee. • Similarly, producers in countries not affected by climatic problems should benefit from increased sales and revenue. Up to 3 marks for explanation/analysis of the extent to which coffee bean producers will not gain: • If the demand for coffee is price elastic in demand, then coffee producers would not be expected to gain (in terms of reduced revenue). • For example, those producers unable to produce a sufficiently large enough yield may make losses which may ultimately result in business closures. • Similarly, depending on the strength of the XED, total revenue may fall if the huge price increase in 2021 results in sufficient numbers of consumers switching their preferences to substitute goods e.g. tea. Up to 3 marks maximum for each perspective, with an overall maximum of 4 marks. Up to 2 marks for evaluation: • That clearly considers whether or not coffee bean producers will gain from the huge increase in coffee bean prices in 2021. • Comes to a reasoned conclusion as to whether advantages outweigh disadvantages or vice versa (reserve 1 mark) Note: No mark for eval can be awarded if only one perspective considered.	6

Case Study 2

Question	Answer	Marks
2	Using the concept of price elasticity of demand, assess the relative impact on poorer households, including those on fixed incomes, of rising prices of food and energy. For explanation/ analysis (up to 4 marks) For a fully accurate formula of PED and / or an explanation of what it measures in terms of the responsiveness / percentage change in quantity demanded compared to a change in price (1) For an explanation that food and energy would be classed as essential/necessity goods and therefore price inelastic meaning that quantity demanded would be less responsive to a change in price / the fall in quantity demanded would be less than the increase in price (1) For an explanation that poorer households (including fixed income households) would suffer a great detrimental / negative effect because their income does not rise in line with inflation and / or they are unlikely to be in a position to negotiate for higher wages (1) For an explanation that this situation can only be overcome if e.g., government support is made available to assist poorer households with the rising prices of food and energy. (1) ALTERNATIVELY, the 4th mark can be given for a comparison to higher income households who will probably be in a position to either negotiate for income increases in line with inflation OR absorb the rising prices of food and energy. For evaluation That clearly assesses the extent of the impact solely on lower income households e.g., by considering the amount of possible government assistance possible, the degree to which such households can compensate for the price rises etc., Reserve 1 mark for a justified conclusion. OR That clearly assesses the relative impact on different types of households (including high-income households) by considering e.g., the proportion of income spent on food and energy together with their ability to negotiate for higher pay. Reserve 1 mark for a justified conclusion. Please use a text box to show the mark split e.g., 3,1	6

Section B

Question	Answer	Marks
1	With the help of a formula, explain what is meant by the term price elasticity of demand for a product <u>and</u> consider the extent to which it allows an entrepreneur to determine the effect of price changes on the total expenditure on their product. Follow the point-based marking guidance at the top of this mark scheme and award: • up to 3 marks for AO1 Knowledge and understanding • up to 3 marks for AO2 Analysis • up to 2 marks for AO3 Evaluation. AO1 Knowledge and Understanding (max 3 marks) For a clear understanding of the meaning of price elasticity of demand (1) plus an accurate formula (1) plus a clear understanding of what is meant by price elastic and price inelastic that refers to relative proportionate or percentage changes rather than large or small changes (1) AO2 Analysis (max 3 marks) Up to 2 marks for clear analysis of how total expenditure on a product is affected by price changes if the demand is price elastic. Up to 2 marks for clear analysis of how total expenditure on a product is affected by price changes if the demand is price inelastic. Note: as the question refers to price changes it is expected that answers should consider both price increases and price decreases. If either part of the analysis only considers a price rise or a fall, then a maximum of 1 mark for that part of the analysis. If both parts of the analysis consider a price rise or fall then a maximum mark of 2 i.e., 1 + 1. AO3 Evaluation (max 2 marks) Up to 2 marks for considering the extent to which the concept can help the entrepreneur and making a judgement. • Consideration of the accuracy of the measure and the ease of calculation for an entrepreneur • Consideration of other factors which may affect total expenditure on a product when price changes. • Reserve 1 mark for a conclusion that focuses on the extent to which it allows the entrepreneur to determine the effect of price changes.	8
	AO1 Knowledge and understanding	3
	AO2 Analysis	3
	AO3 Evaluation	2

Section C

1.	C	8.	C	15.	D
2.	C	9.	D	16.	D
3.	D	10.	D	17.	D
4.	A	11.	D	18.	B
5.	A	12.	C	19.	D
6.	D	13.	C	20.	B
7.	B	14.	B		